



Project 20 Test-Retest Reliability of Joint Space Width Measurements from Knee Radiographs

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1. SAS dataset

Name: **kXR_QJSW_Rel_DuryeaXX** (xx identifies the time point)

Display label: **kXR SQ reliability readings (Duryea)**

2. Purpose and methods

2.1 Background

Project 16 (ARRA Radiologic Outcomes for the OAI) consists of quantitative joint space width (JSW) measurements, and related parameters, from knee radiographs done in the laboratory of Dr. Jeff Duryea at Brigham and Women's Hospital in Boston, MA. To evaluate test-retest reliability, images for a subset of participants were resent to the reading center after the original readings were completed. The reading center was blinded to the fact that these were retest readings and blinded to the original JSW measurements. Both original and retest readings are included in the kXR_QJSW_Rel_DuryeaXX datasets, and are denoted by their project number in the READPRJ variable. Records with READPRJ=20A are the original Project 16 readings (which can also be found in the kXR_QJSW_DuryeaXX datasets that contain all the Project 16 ARRA data), records with READPRJ=20B are the retest readings for Project 16, records with READPRJ=20C are the original readings from Project 05, and records with READPRJ=20D are the retest readings for Project 05. Project 05 readings have been removed from the kXR_QJSW_DuryeaXX datasets, but a subset of Project 05 readings are included in the kXR_QJSW_Rel_DuryeaXX datasets since retest data is available for these records. Project 20D data replace the reliability data previously released (Project 13).

2.2 Reliability at single time points

To test reliability of cross-sectional measurements, we compared the baseline visit values from the original reading and the matching retest reading for a subset of the JSW measurements performed for both minimum JSW in the medial compartment (variable V00MCMJSW), as well as fixed location JSW in the medial compartment (variables V00JSW150, V00JSW200, V00JSW250 and V00JSW300), and fixed location JSW in the lateral compartment (variables V00LJSW700, V00LJSW750, V00LJSW800, V00LJSW850). For each knee, this involved pairing a measurement from READPRJ=20A with the matching measurement from READPRJ=20B and similar pairing for 20C and 20D. Bland-Altman plots were calculated along with limits of agreement¹, and intraclass correlation coefficients (ICCs)² were also estimated.

2.3 Longitudinal reliability

For longitudinal reliability, differences between follow-up visit JSW measurements and baseline visit JSW measurements were calculated for each of the separate reading project (20A, 20B, 20C, and 20D) by merging together the kXR_QJSW_DuryeaXX datasets by ID, SIDE and READPRJ, and then the reliability of the change between baseline and follow up was analyzed similarly to the cross-sectional measurements using Bland-Altman plots, limits of agreement, and ICCs.



3. Results

Cross-sectional reliability

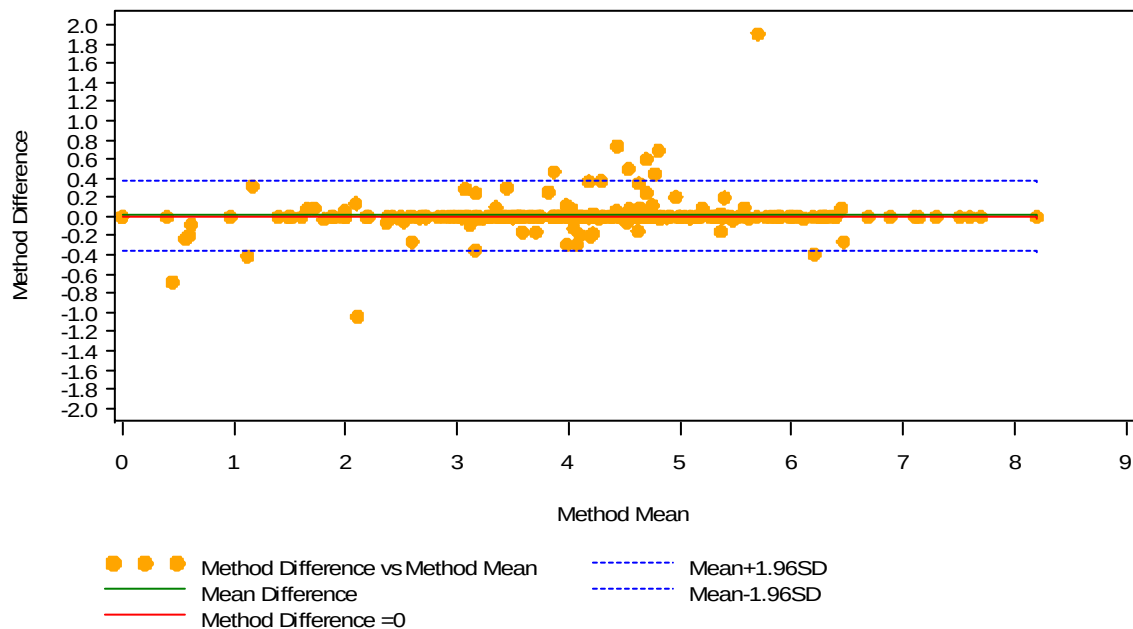
Cross-sectionally, reliability of JSW measurements was high (see Table 1), with ICCs greater than 0.96 for all measurements. Limits of agreement (95% confidence limits on individual measurements) were also good in the medial compartment where values were reliable to ± 0.4 mm or less, apart from V00JSW300. In the lateral compartment, even though ICCs were high, the measurements were not as reliable. The most reliable lateral compartment JSW variable was V00LJSW750.

Table 1. Reliability of baseline visit JSW measurements.

Variable		ICC	Mean (SD) Difference between original & repeat (mm)	Limits of Agreement (mm)	
				Min	Max
V00MCMJSW	medial minimum	0.984	0.014 (0.185)	-0.349	0.377
V00JSW150	medial (x=0.15)	0.986	-0.014 (0.186)	-0.379	0.351
V00JSW200	medial (x=0.20)	0.987	-0.009 (0.168)	-0.339	0.319
V00JSW250	medial (x=0.25)	0.982	-0.002 (0.192)	-0.397	0.375
V00JSW300	medial (x=0.30)	0.964	-0.008 (0.296)	-0.589	0.572
V00LJSW700	lateral (x=0.70)	0.961	0.054 (0.530)	-0.983	1.092
V00LJSW750	lateral (x=0.75)	0.984	-0.078 (0.309)	-0.682	0.527
V00LJSW800	lateral (x=0.80)	0.961	-0.076 (0.497)	-1.050	0.899
V00LJSW850	lateral (x=0.85)	0.967	-0.041 (0.448)	-0.919	0.837

An example Bland-Altman Plot for minimum JSW is shown below (Figure 1).

Figure 1. A Bland-Altman plot for test-retest measurement of V00MCMJSW. The limits of agreement are shown by the distance between the two horizontal dashed lines.





Longitudinal reliability

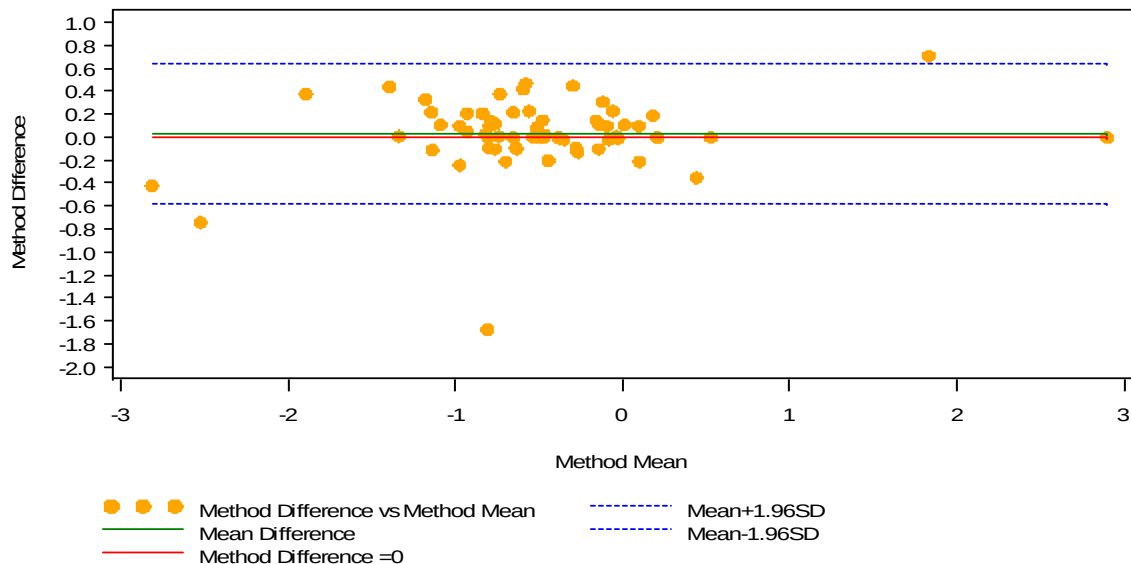
Measurements of the change in JSW from baseline to 36-month had good reliability. ICCs for change were in the range 0.93 to 0.96 medially, and slightly lower laterally. Limits of agreement appeared good for medial fixed location JSW, and was best for positions $x=0.15$ and $x=0.20$ at around ± 0.4 mm. Laterally, reliability for measuring change was not as good, although JSW at $x=0.80$ seemed the most reliable location for measuring change.

Table 2. Reliability of change from baseline to 36-month visit for JSW measurements.

Change Variable (V05xxx – V00xxx)		ICC	Mean (SD) Difference between original & repeat (mm)	Limits of Agreement (mm)	
				Min	Max
MCMJSW	medial minimum	0.955	-0.060 (0.235)	-0.522	0.401
JSW150	medial ($x=0.15$)	0.930	-0.014 (0.186)	-0.379	0.351
JSW200	medial ($x=0.20$)	0.953	-0.002 (0.192)	-0.379	0.375
JSW250	medial ($x=0.25$)	0.932	-0.044 (0.303)	-0.637	0.550
JSW300	medial ($x=0.30$)	0.950	0.019 (0.310)	-0.590	0.628
LJSW700	lateral ($x=0.70$)	0.898	-0.056 (0.435)	-0.909	0.798
LJSW750	lateral ($x=0.75$)	0.888	-0.033 (0.432)	-0.879	0.813
LJSW800	lateral ($x=0.80$)	0.925	0.032 (0.309)	-0.574	0.638
LJSW850	lateral ($x=0.85$)	0.852	0.002 (0.447)	-0.873	0.878

Results for baseline to 24-month, and baseline to 12-month change were similar, suggesting that length of follow-up does not increase reliability for measuring change. An example Bland-Altman plot for the longitudinal change of lateral JSW at $x=0.8$ is shown below in Figure 2.

Figure 2. A Bland-Altman plot for test-retest measurement of (V05LJSW800-V00LJSW800). The limits of agreement are shown by the distance between the two horizontal dashed lines.





4. References

1. Bland JM, Altman DG Statistical methods for assessing agreement between two methods of clinical measurement. *Lancet* 1986, 327 (8476): 307–10.
2. Shrout, P. and Fleiss, J. L Intraclass correlation: uses in assessing rater reliability. *Psychological Bulletin*. 1979, Vol. 86, No. 2, pp. 420–428.